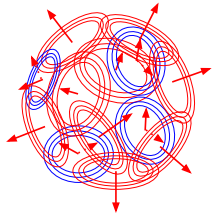
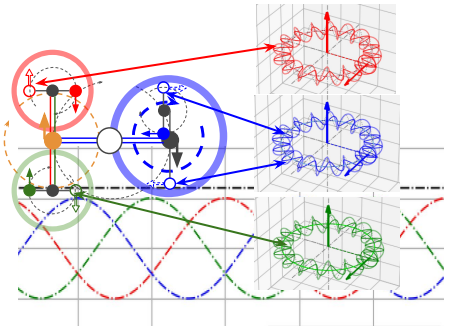
R-space Theory 1.0

- Geometric Integration of Relativity and Quantum Mechanics -



Chap.7:
The Mechanism of Color Confinement



1. What are Quarks?

In the 1930s, it was believed that the universe was composed of three basic particles: electrons (*e*−), protons (*p*), and neutrons (*n*).

However, from the late 1950s, the performance of accelerators and detectors improved dramatically, leading to the discovery of a large number of new particles known as hadrons (particles like protons and neutrons that undergo strong interactions).

By the early 1960s, more than 100 types of hadrons had been identified.



1. What are Quarks?

In 1964, Murray Gell-Mann and George Zweig independently proposed a model explaining these hadrons systematically, stating that "hadrons are made of three basic particles".

Gell-Mann named these fundamental particles "Quarks".

Later, during deep inelastic scattering experiments at SLAC (Stanford Linear Accelerator Center) in 1968–69, colliding high-energy electrons with protons revealed the existence of "point-like cores" (partons) inside protons.

These partons were eventually identified as quarks and gluons, providing evidence that hadrons such as protons and neutrons are indeed composed of quarks.

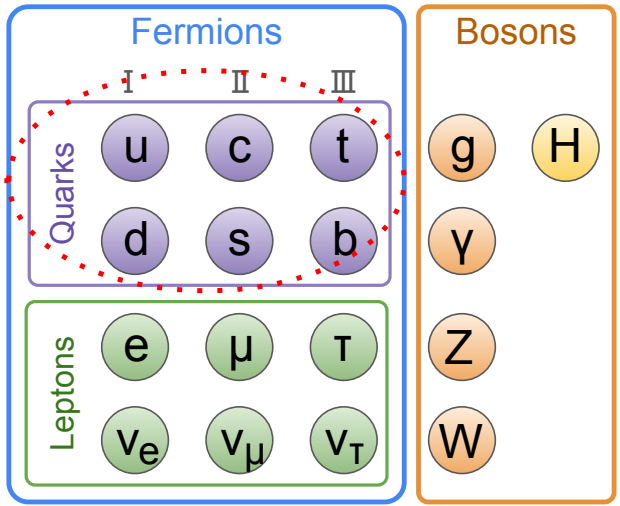
1. What are Quarks?

In the Standard Model of quantum mechanics, there are six types of quarks, and it is believed there are a total of 17 types of elementary particles, excluding the yet-to-be-discovered graviton.

Hadrons in the quark model

p (Proton)	uud
n (Neutron)	udd
Λ (Lambda)	uds
Σ (Sigma)	uus, uds, dds
Ξ (Xi)	uss, dss
Δ (Delta)	uuu, uud, udd, ddd
Λc (Lambda-c)	udc
…	

Standard Model of particle physics

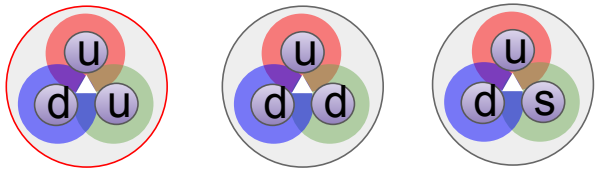


1. What are Quarks?

Among hadrons, those composed of three quarks (such as protons and neutrons) are classified as baryons, while those composed of two quarks (such as pions) are classified as mesons.

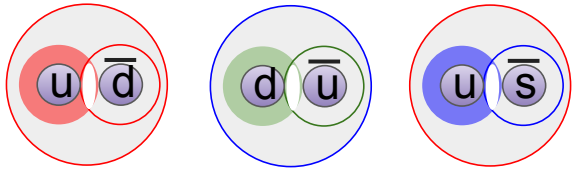
There are also particles called exotic hadrons, such as tetraquarks consisting of four quarks, which are observed only in accelerator experiments and have very short lifespans.

Baryons



Proton, Neutrino, Lambda…

Mesons

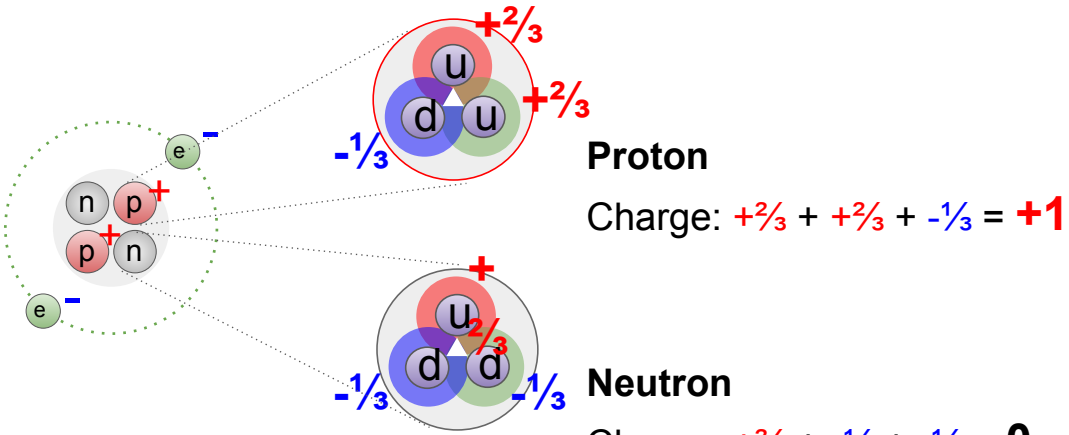


π+, π-, k+...

1. What are Quarks?

In contrast to the negative 1 charge of an electron, quarks possess fractional charges. An up quark has a +2/3 charge, and a down quark has a −1/3 charge.

A proton, consisting of two up quarks and one down quark (u,u,d), has a total charge of +1, while a neutron, consisting of one up quark and two down quarks (u,d,d), has a total charge of zero.



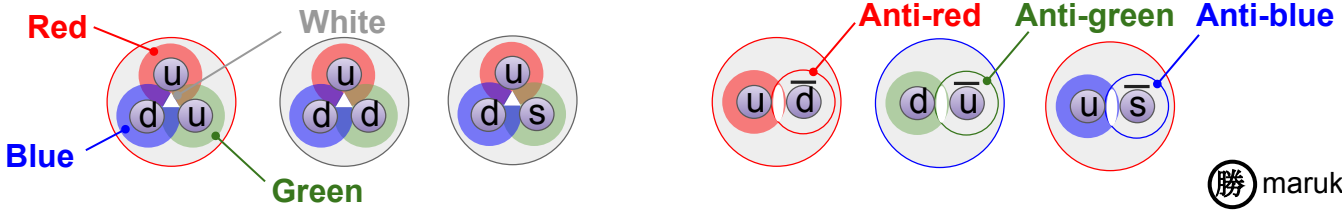
1. What are Quarks?

Quarks possess the mysterious property of being unable to exist as isolated elementary particles; they only exist as constituents of hadrons.

This principle is known as color confinement, where quarks are said to possess color charges: red, green, and blue (or anti-red, anti-green, and anti-blue).

In hadrons like protons and neutrons, these three color charges neutralize to become "white," causing the quarks to be confined.

(Note: Quarks do not actually have colors; this is an analogy based on how the primary colors of light combine to form white light).

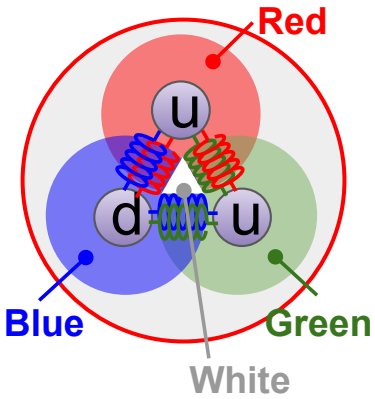


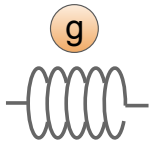
1. What are Quarks?

The strong force, which acts as the interaction between quarks for color confinement, is mediated by gluons, which also carry color charge.

While the Coulomb force from electric charges weakens inversely with the square of the distance, gluons behave like a spring, where the force increases in proportion to the distance.

Color confinement and gluon





gluon

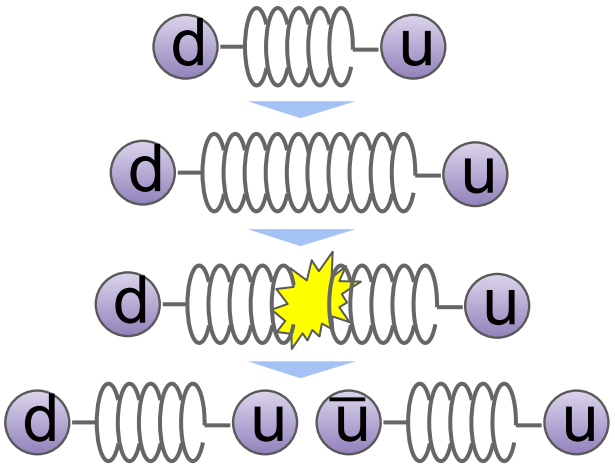
Eight gluon color states

- red–anti-blue gluon
- blue–anti-green gluon
- green–anti-red gluon
- …

1. What are Quarks?

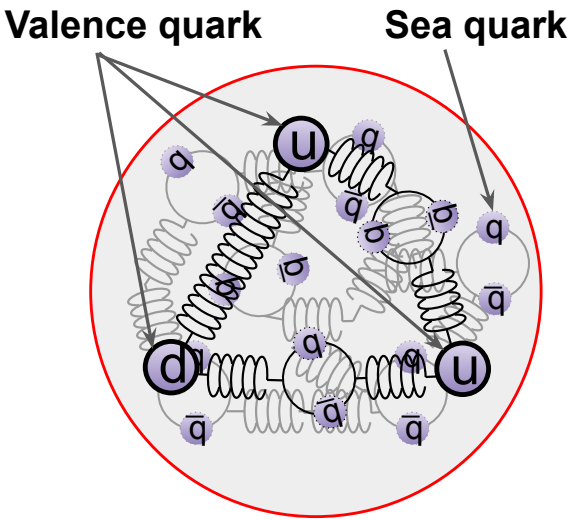
Therefore, if quarks move apart, the energy of the gluons increases, and that energy generates new quarks, resulting in the creation of other elementary particles.

This prevents quarks from existing in isolation.



1. What are Quarks?

Additionally, it is now believed that inside protons and neutrons, there exist "sea quarks"—temporary quarks that undergo repeated pair production and annihilation from gluon energy—in addition to the three "valence quarks" .



2. The Mechanism of Fractional Charges

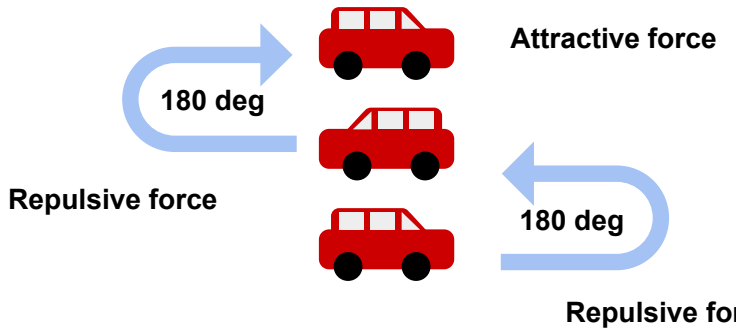
As explained in Chapter 2, in R-space Theory, the number of rotation vectors representing angular momentum determines the electric charge, where one rotation vector equals a −1/3 charge.

A positive charge acts as an attractive force (360 degrees = 180 degrees + 180 degrees) by rotating the repulsive force of a negative charge (inversion = 180-degree rotation).

$$q = (-1)^n \cdot \frac{n}{3}$$

q: charge

n: number of rotation vectors

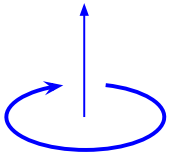


2. The Mechanism of Fractional Charges

A down quark has one rotation vector, resulting in a −1/3 charge.

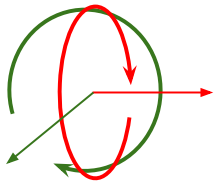
An up quark has two rotation vectors, resulting in a +2/3 charge.

$$q = (-1)^n \cdot \frac{n}{3}$$



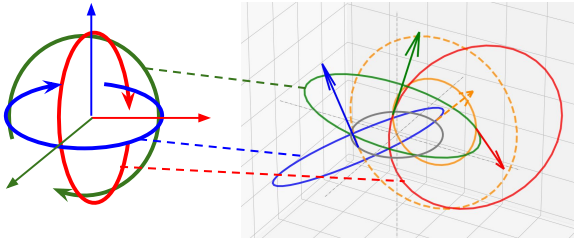
down quark

q = (-1)^**1** * **1**/3 = **-1**/3



up quark

q = (-1)^**2** * **2**/3 = **+2**/3



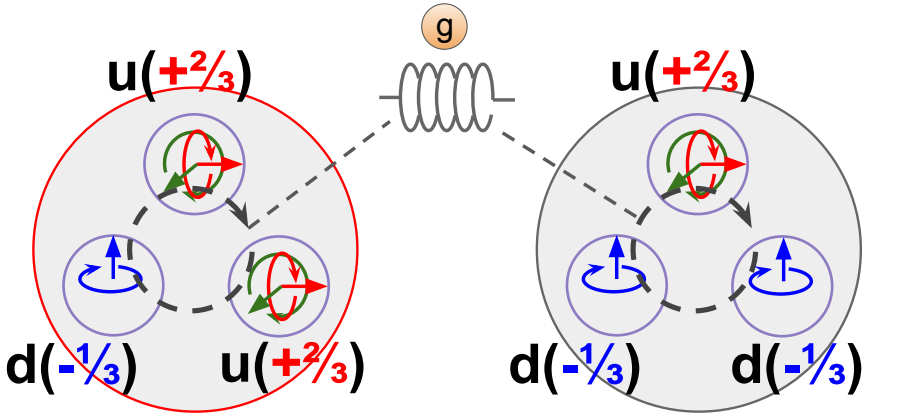
Electron

q = (-1)^**3** * **3**/3 = **-3**/3 = -1

2. The Mechanism of Fractional Charges

When protons (u,u,d) and neutrons (u,d,d) are represented by rotation vectors, they appear as shown in Figure.

The force that binds (confines) these quarks is the gluon, which is related to another rotation mode inherent in the rotation vectors.



Proton

Charge: +⅔ + +⅔ + -⅓ = **+1**

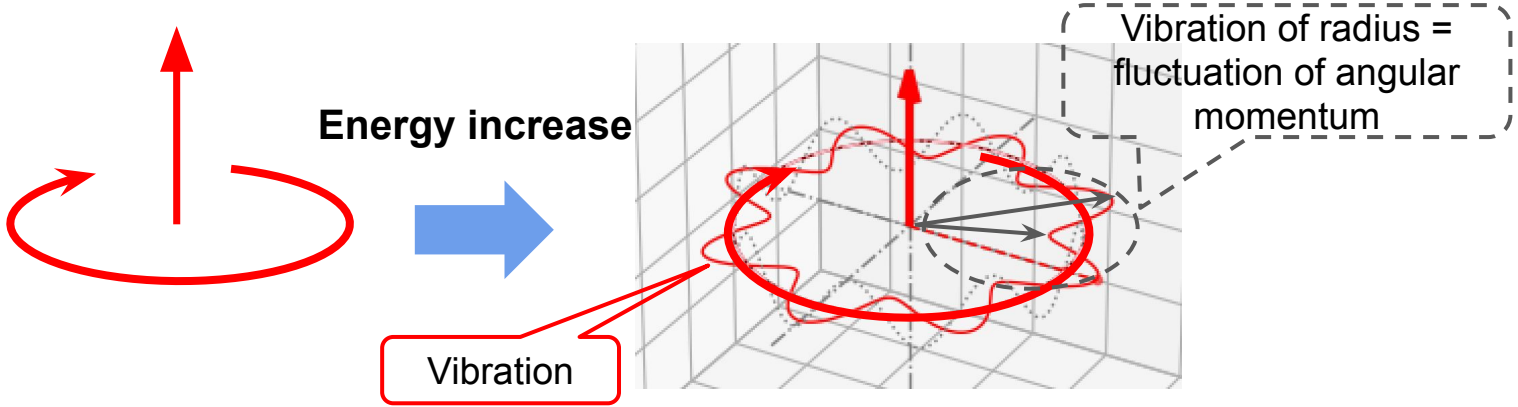
Neutron

Charge: +⅔ + -⅓ + -⅓ = **0**

3. The Mechanism of Color Charge

As the energy of a rotation vector increases, the phase circle of the rotation vector begins to vibrate like a string.

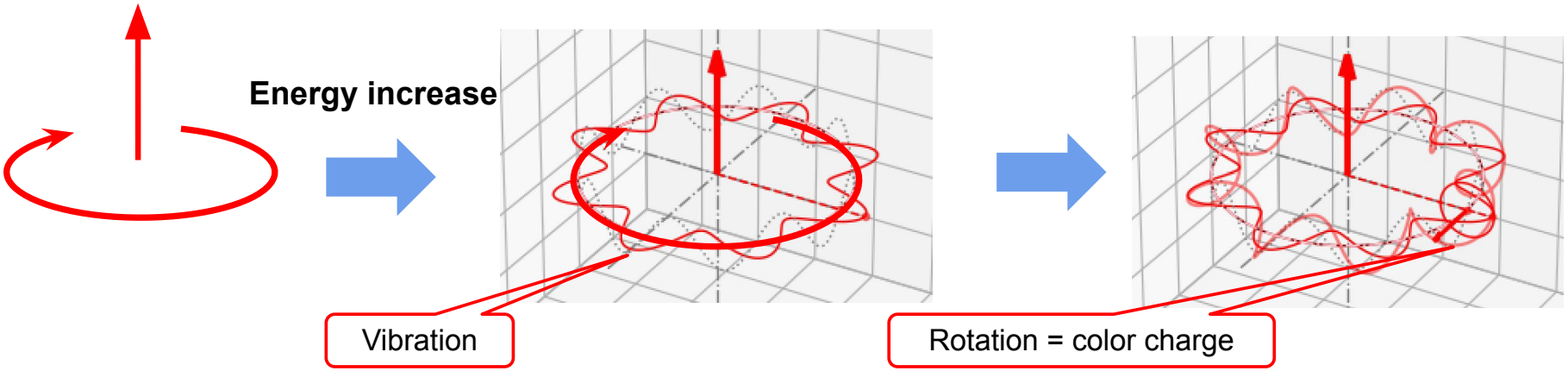
Since the size (radius) of the phase circle represents the magnitude of angular momentum, this vibration represents the fluctuation of angular momentum.



3. The Mechanism of Color Charge

If the direction of this fluctuation (large to small, small to large) is taken along the axis of the rotation vector, the vibration of angular momentum can be thought of as a torus-shaped rotation.

This vibration of angular momentum is the color charge.

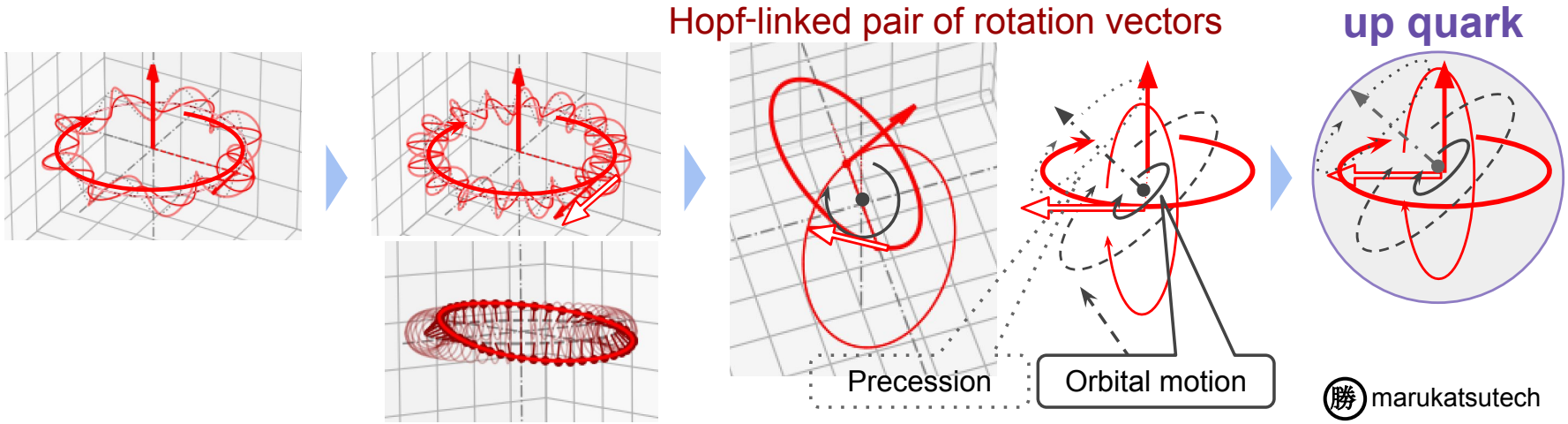


3. The Mechanism of Color Charge

As the vibration frequency increases, the rotation vector (angular momentum) of the color charge grows.

It then balances with the main rotation vector to form a Hopf-link-shaped pair, performing orbital and precessional motion.

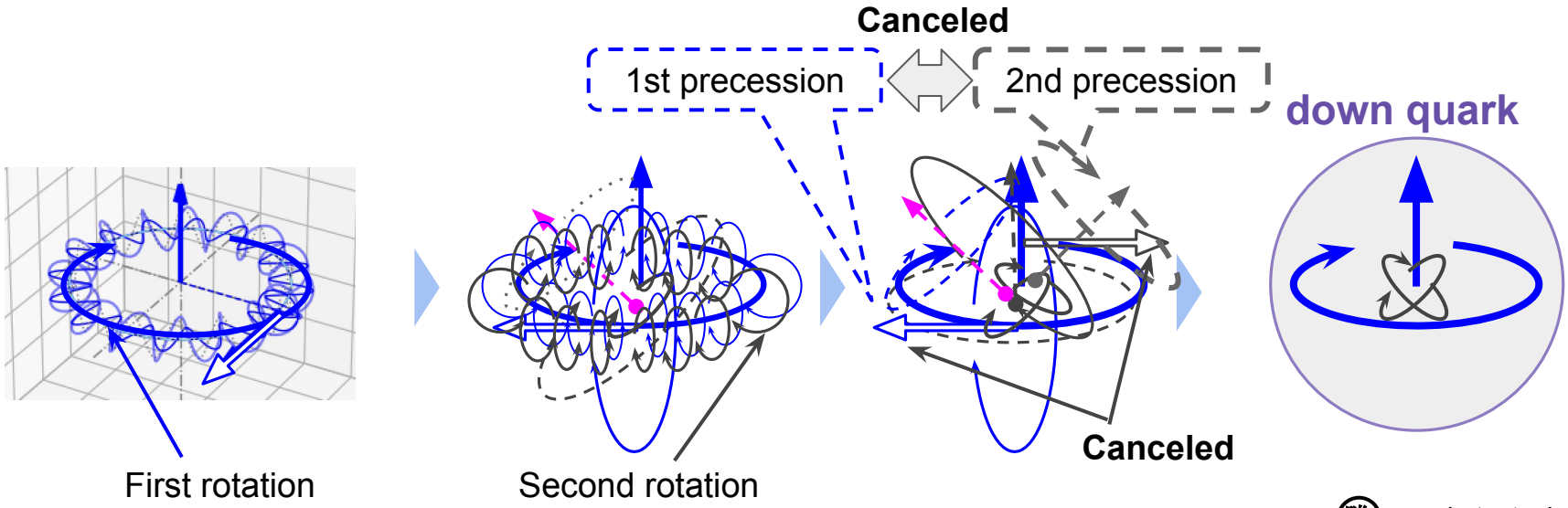
This is the up quark.



3. The Mechanism of Color Charge

A down quark appears to be a simple single rotation vector without orbital or precessional motion, but its rotation vector is also vibrating.

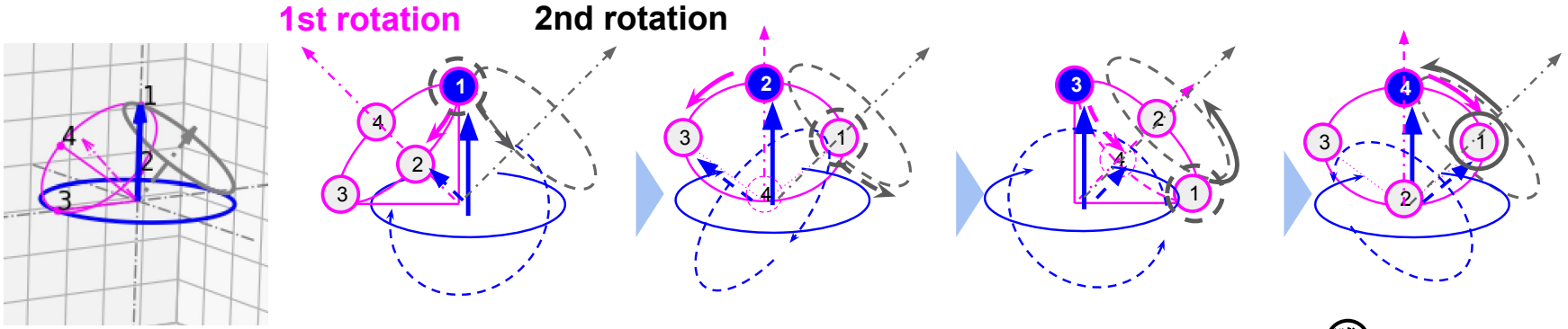
It vibrates at twice the frequency of an up quark.



3. The Mechanism of Color Charge

Because it rotates twice, it undergoes orbital/precessional motion during the first rotation (achieving an up-quark-like state), but this motion is canceled out during the second rotation, making it appear as though no such motion is occurring.

This is why the mass of a down quark is larger than that of an up quark (as it possesses twice the vibration energy).



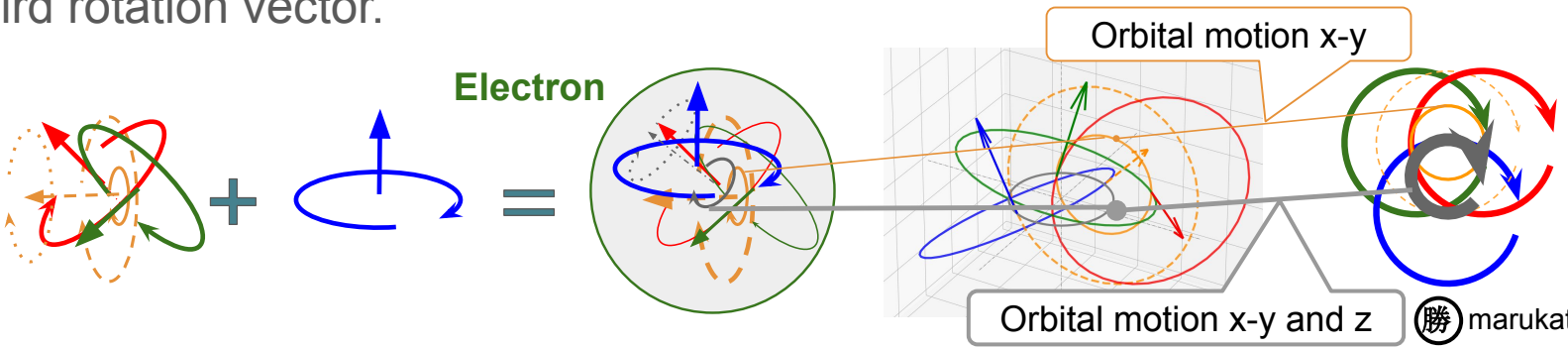
4. Color Confinement and the Proton Model

The structure of a proton is similar to that of an electron.

A proton is essentially formed by the rotation vectors of an electron being flipped (rotated) by the vibration of its angular momentum to become quarks.

As explained in Chapter 2, the structure of an electron involves two rotation vectors forming a Hopf-link pair that performs orbital and precessional motion.

The electron is formed when this pair's motion further forms a Hopf-link with a third rotation vector.



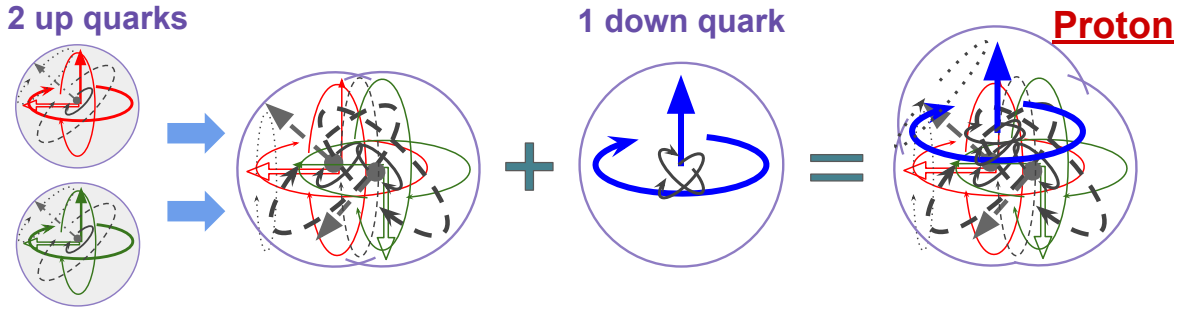
4. Color Confinement and the Proton Model

In the case of a proton, the up and down quarks (composed of rotation vectors as described in Section 3) form Hopf-link pairs.

First, two up quarks pair up.

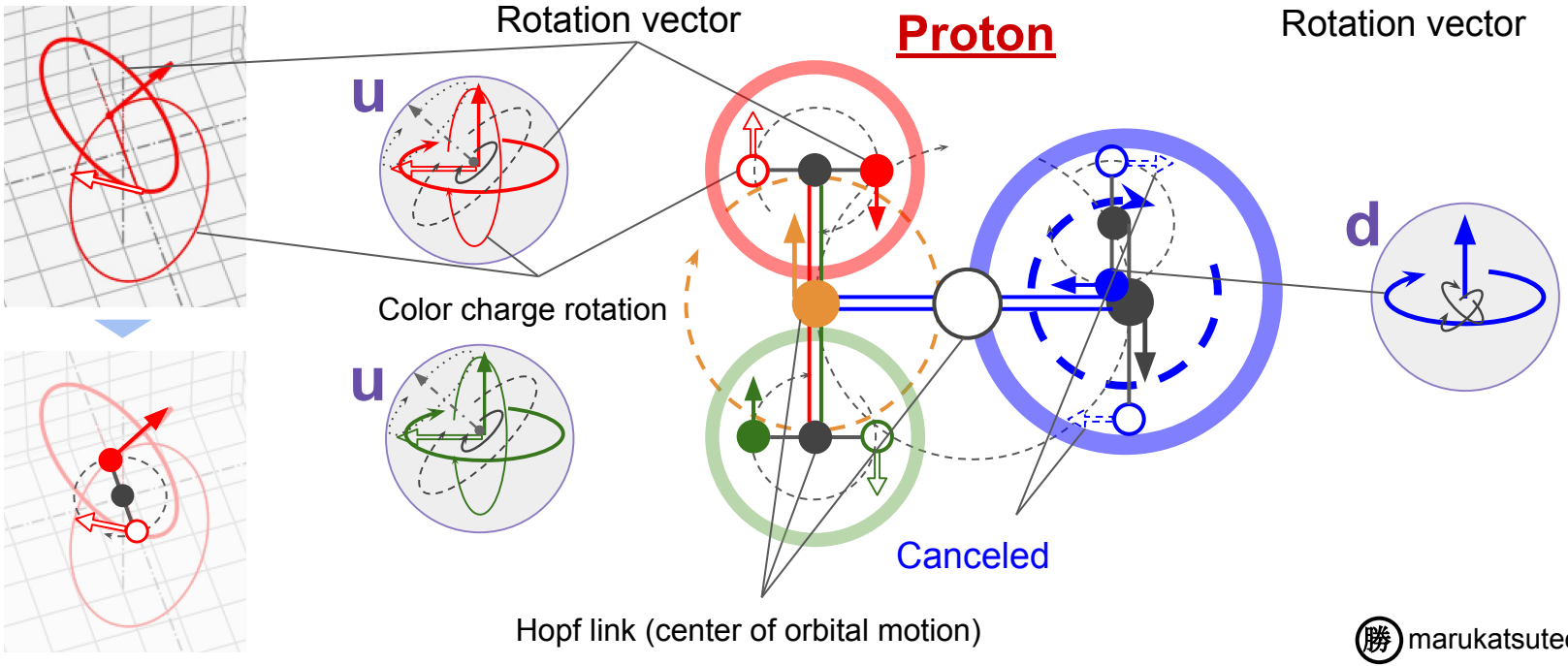
Since both up quarks are performing orbital/precessional motion, their respective rotations form a Hopf-link pair as a collective rotation vector.

The proton is then formed by this Hopf-link pair further forming another Hopf-link with a down quark.



4. Color Confinement and the Proton Model

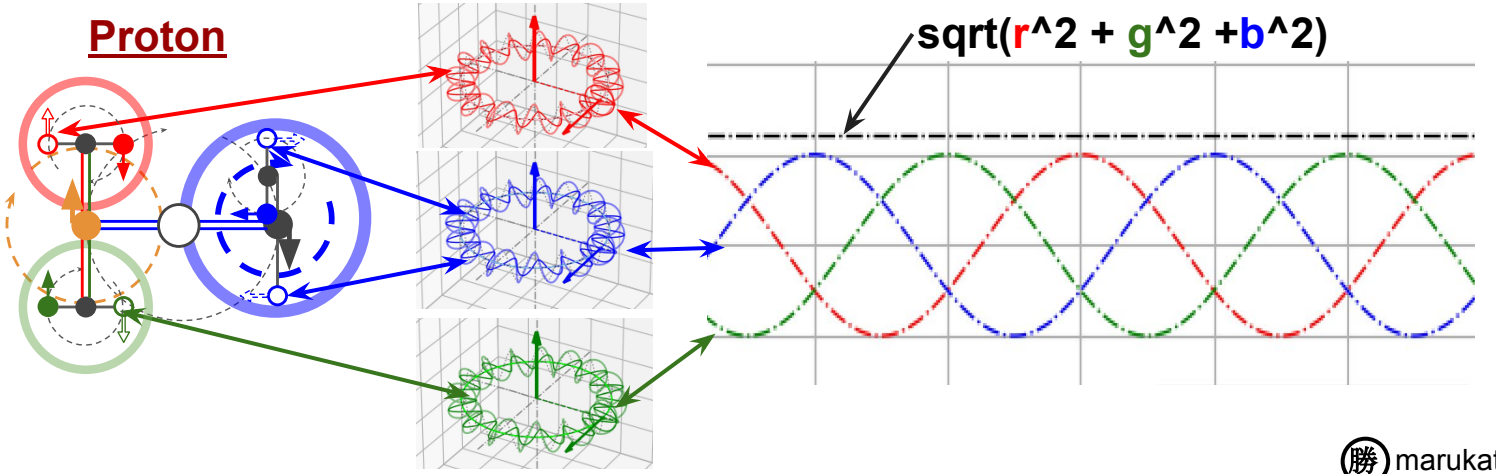
To simplify the relationship between the phase circles and the orbital motion of the Hopf-links, they can be illustrated in a planar diagram.



4. Color Confinement and the Proton Model

The three rotation vectors (x: red, y: green, z: blue) do not vibrate randomly; they cancel each other out and maintain balance like three-phase alternating current.

This is the meaning behind the analogy that color confinement is like the primary colors of light combining to form white light.

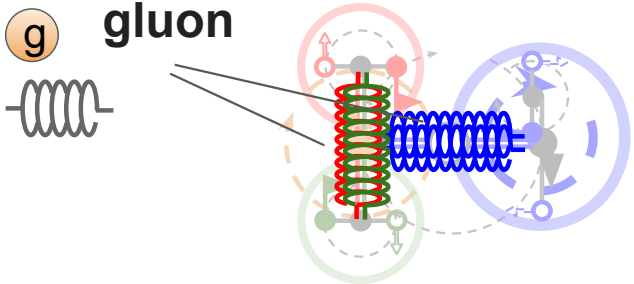


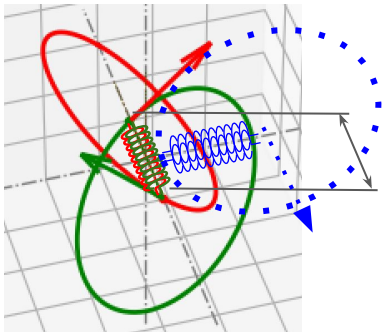
4. Color Confinement and the Proton Model

Furthermore, the vibration of angular momentum is also the vibration of the distance between the rotation vectors in the Hopf-link.

If the angular momentum becomes too large, the Hopf-link cannot be maintained, and the energy causes the pair of rotation vectors to split.

They then pair with new rotation vectors produced through pair production from that energy, transforming into other particles (quanta). This spring-like property between the Hopf-links is the gluon.





Distance between the rotation vector

4. Color Confinement and the Proton Model

This spring-like property originates from the action of trying to stabilize the vortex of angular momentum.

This action stems from the symmetry of the R-aether, the medium of the angular momentum vortex.

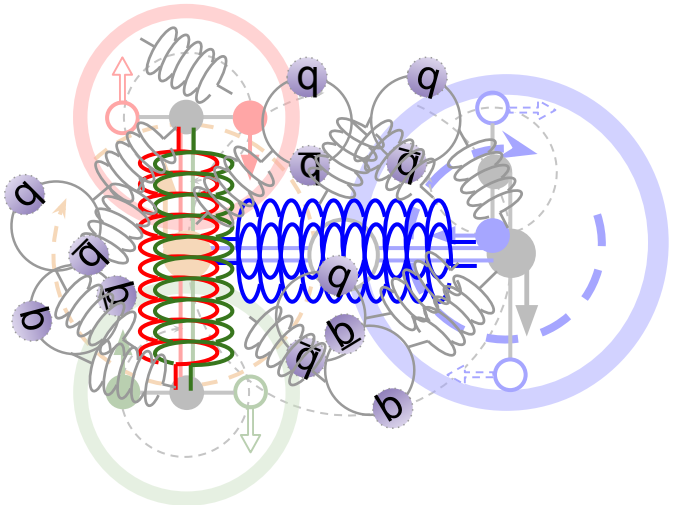
As explained in later chapters, conservation laws in physics arise from symmetry, as described by Noether's theorem.

Quarks maintain balance in color confinement because the R-aether preserves symmetry and seeks to minimize action.

4. Color Confinement and the Proton Model

Additionally, the sea quarks are the result of quantized fluctuations in the vibration energy between the Hopf-links (caused by angular momentum vibration), leading to repeated pair production and annihilation.

Sea quarks



5. The Mechanism of Neutrons and Beta Decay

Neutrons are slightly more complex than protons.

A neutron consists of one up quark and two down quarks.

If one considers a simple pairing of three quarks like a proton, the mass (energy) balance is poor.

The mass of an up quark is approximately 2.2 MeV/c^2, while a down quark is approximately 4.7 MeV/c^2.

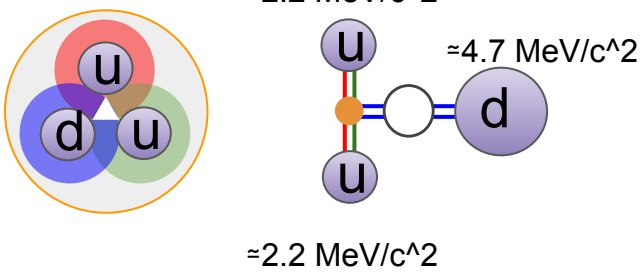


5. The Mechanism of Neutrons and Beta Decay

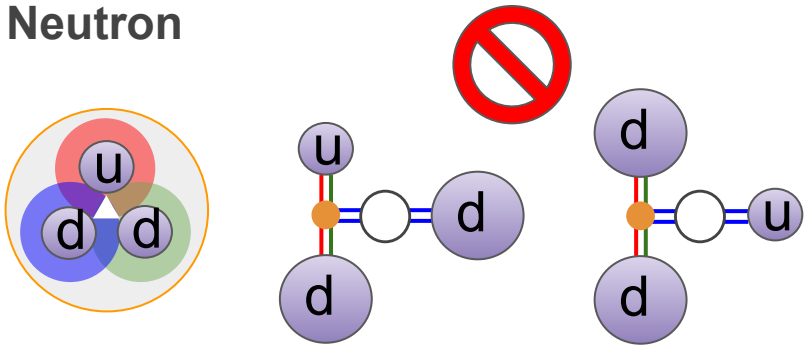
If an up and down quark form the first pair, the mass balance is not maintained.

Similarly, if two down quarks form the first pair, there is a large mass difference with the remaining up quark.

Proton



Neutron

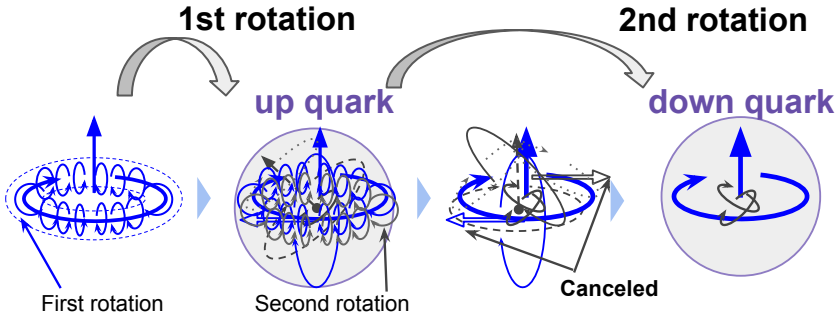


5. The Mechanism of Neutrons and Beta Decay

A neutron is formed when additional energy is added to a proton, causing the three rotation vectors (quarks) to flip (rotate) further due to angular momentum vibration, thereby changing the quark types.

As explained in Section 4, a rotation vector becomes an up quark when orbital/precessional rotation is added via vibration.

With further vibration, the first orbital/precessional motion is canceled, resulting in a down quark.

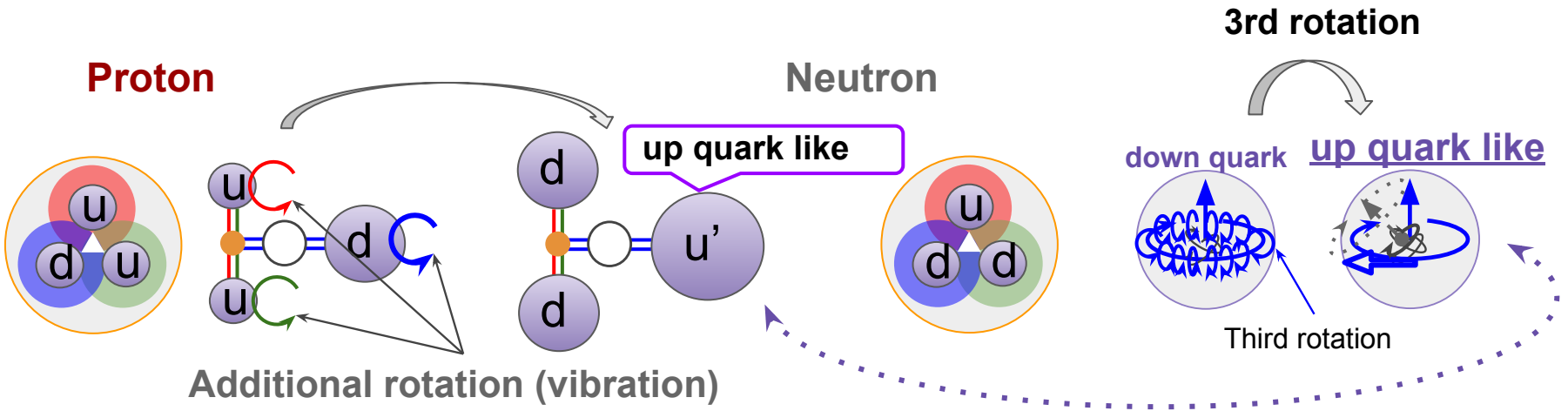


5. The Mechanism of Neutrons and Beta Decay

In this process, both up quarks in a proton change into down quarks.

Then, rotation is added to a down quark, causing it to begin orbital/precessional motion and enter an up-quark-like state.

This is the neutron.

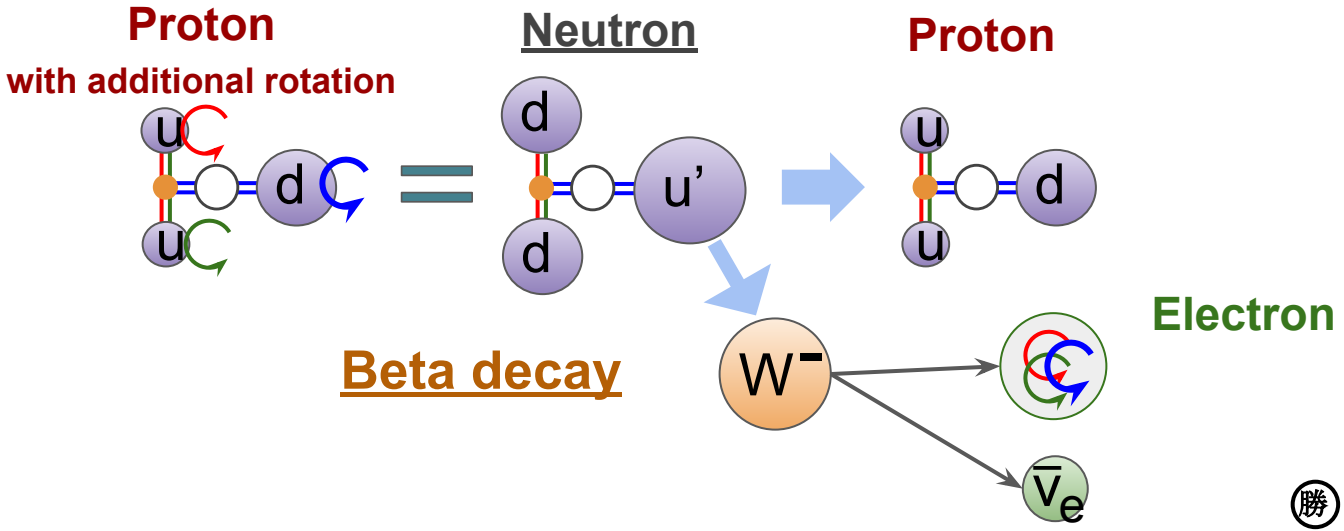


5. The Mechanism of Neutrons and Beta Decay

Because this neutron state is unstable, it returns to a proton by releasing the added rotational energy as a W-boson.

The W-boson then decays into an electron and an anti-electron neutrino.

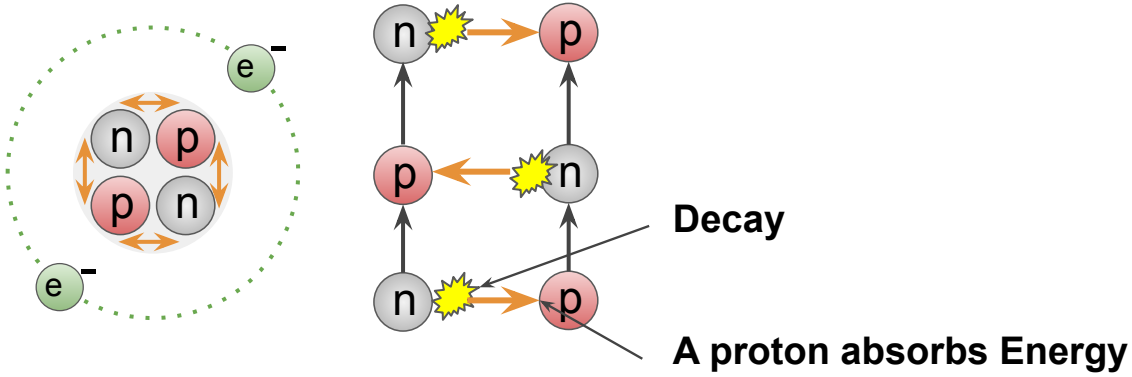
This is beta decay.



5. The Mechanism of Neutrons and Beta Decay

While neutrons appear stable inside an atomic nucleus, they are constantly undergoing beta decay.

A neutron decays, a proton absorbs that energy to become a neutron, and the cycle repeats.



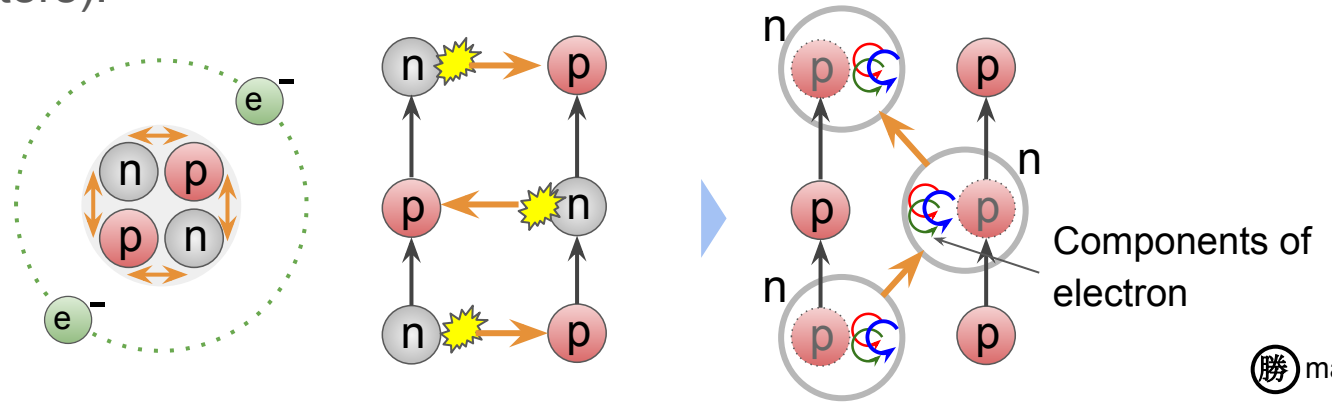
5. The Mechanism of Neutrons and Beta Decay

In a sense, one could say that everything inside a nucleus is a proton.

Protons are exchanging vibrational (rotational) energy equivalent to electrons.

It is the electron component that attracts the protons to each other, allowing the nucleus to exist stably.

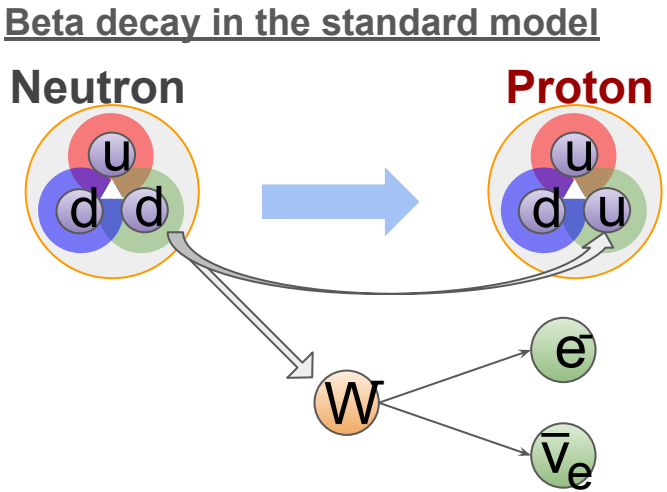
(The mechanisms of W-bosons and anti-electron neutrinos will be explained in later chapters).



5. The Mechanism of Neutrons and Beta Decay

Note: In the Standard Model, the force binding protons and neutrons is attributed to residual interactions leaking from the gluon force or the exchange of pions.

Also, beta decay is described as one of the two down quarks in a neutron changing into an up quark.

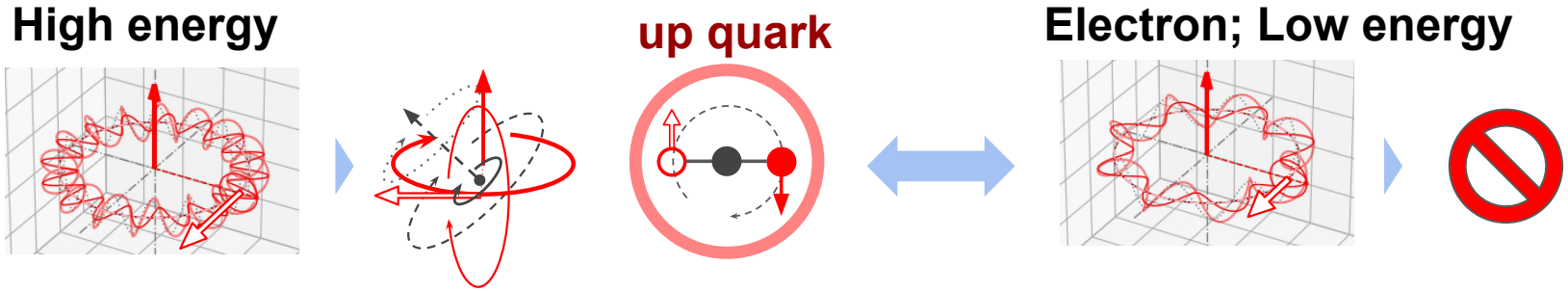


6. Angular Momentum Vortices and Particles

As explained above, quarks, like photons and electrons, are composed of vortices of angular momentum.

Electrons also possess color charge.

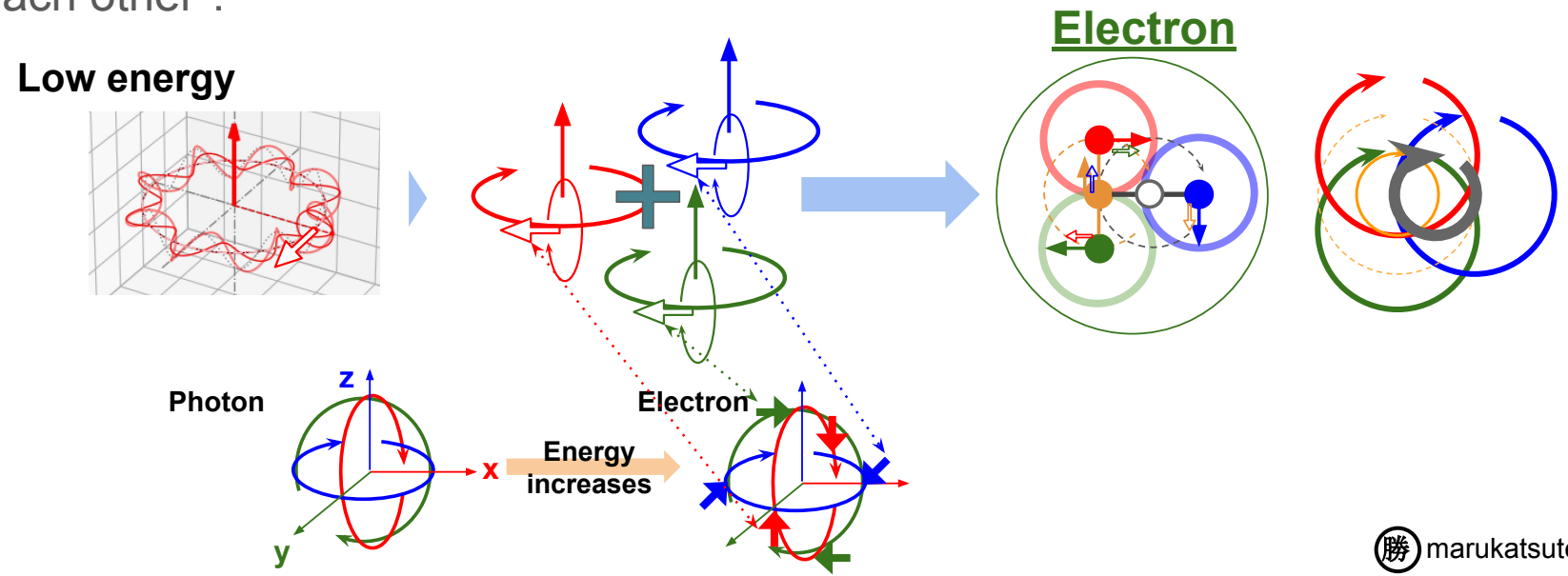
However, they have less energy (angular momentum vibration) than quark color charges and cannot perform orbital/precessional motion with a single rotation vector.



6. Angular Momentum Vortices and Particles

Therefore, they pair with the three rotation vectors (x, y, z) to perform orbital and precessional motion.

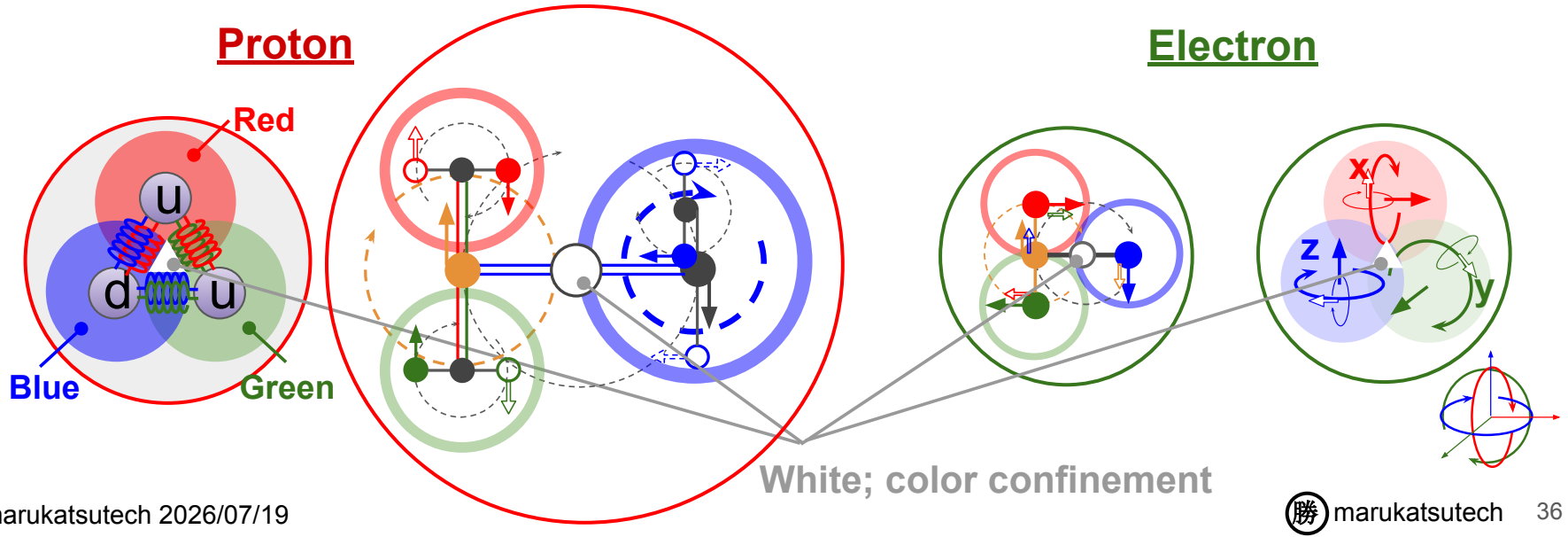
This is the mechanism described in Chapter 2 where "rotation vectors influence each other".



6. Angular Momentum Vortices and Particles

This is the mechanism by which the angular momentum of a photon rotates and becomes confined as an electron.

(The generations of elementary particles will be explained in the next chapter).



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This is all for my presentation.

Thanks for watching!

